

Rain Liao

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[Linkedin](#) | [Project portfolio](#) | [Github](#)

PERSONAL SUMMARY

Versatile Technical Professional with extensive experience in cloud-based system deployment on AWS and Azure, specialising in Linux-based operations and REST API integrations. Proven ability to troubleshoot complex technical environments, including reducing system setup times by 80% through Docker containerisation. Combines deep Python and SQL expertise with strong communication skills to resolve high-priority issues and ensure seamless service delivery.

CERTIFICATION

- Microsoft Azure Data Scientist Associate: DP-100
- Microsoft Power BI Data Analyst Associate: PL-300
- First-author conference publication – [ICRTCIS 2025 \(2026\)](#)
- ALTA 2025 – 5th/150 (ADR detection), +20% accuracy

WORK EXPERIENCE

Aug 2025 - Oct 2025 Data science internship

University of Adelaide, SET Faculty

- Engineered an end-to-end ETL pipeline using Python, Selenium, and Pandas to scrape and structure 60,000+ unstructured posts from mental health forums, enabling large-scale NLP analysis with Mental-BERT models.
- Developed a Transformer-based topic modeling system with network analysis that identified 10 common mental health themes, directly informing supervisor-led research decisions and reducing manual content analysis time by ~60%.
- Collaborated with a psychology student to validate chatbot availability and topic relevance, ensuring clinical accuracy and improving system usability for academic research applications.

Aug 2024 - Mar 2025 Quantitative trading assistant

Australian Institute for Machine Learning (AIML)

- Backtested 10+ systematic trading strategies on 100GB+ of Binance market data using Python and Freqtrade, incorporating ChatGPT-4o sentiment analysis to classify stock-specific news as positive/negative signals for improved trade timing.
- Engineered a Docker-based research environment that reduced setup time by 80%, improved prediction accuracy by 13%, and lowered portfolio volatility by 18% through optimized data pipelines and real-time signal processing.
- Presented weekly performance attribution reports to 2 senior researchers, detailing strategy P&L, sentiment signal impact, and volatility metrics to guide iterative model improvements.

Jul 2023 - Oct 2023 Special case technician

National Centre for Research on Earthquake Engineering

- Scraped and processed real-time weather/seismic data from government meteorological APIs using Python and QGIS, creating precipitation heat maps and structuring data into DynamoDB/S3 with numerical/text separation for scalable querying.
- Developed and deployed a cloud-based disaster alert system integrating REST APIs, Google Maps API, and Python for real-time data processing on AWS (EC2, Lambda), reducing alert latency from 3 minutes to 15 seconds while collaborating with urban planning and statistics interns.
- Coordinated with a cross-functional team of 5 interns from urban planning, statistics, mathematics, and civil engineering to integrate the alert system with third-party road blockage prediction models, enabling proactive infrastructure risk assessment.

Jul 2025 - Present Front of House

Olive (Adelaide Australia)

- Provide excellent guest service through clear communication, teamwork, and quick problem-solving in a fast-paced fine dining environment.

LEADERSHIP

Jan 2023 - Jul 2023 National College Smart Innovation Competition (Taipei Taiwan)

Team Leader

- Led a multidisciplinary team of 4 to develop a full-stack web application using React, Django, and SQL, creating a typhoon lethality prediction system that challenged traditional metrics with

- real-time acceleration analysis.
- Coordinated task allocation and technical decisions using Jira and Google Meet, enabling rapid self-learning of full-stack development and on-time delivery of a functional prototype despite non-technical backgrounds.
- Bridged frontend-backend integration by connecting React components with Django APIs and SQL database, establishing a scalable architecture for future development phases.

TECHNICAL SKILLS

Programming: Python, SQL, R, C++	Web & APIs: HTML, CSS, JavaScript, React, REST APIs
Data & ML: ETL, NLP, Transformers, RAG	Tools: Power BI, QGIS, Selenium, BeautifulSoup
	Cloud & DevOps: AWS (EC2, Lambda, S3), Azure, Docker, Linux, Git

EDUCATION

MASTER OF DATA SCIENCE	BACHELOR of CIVIL ENGINEERING
University of Adelaide Adelaide, SA	Chung Yuan Christian University Taoyuan, Taiwan
Feb 2024 – Dec 2025	Sep 2018 - Jun 2023

LANGUAGE

English (IELTS 7.5)	Mandarin (Native)
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REFERENCES

Dr. Indu Bala (Lecturer)	Bernard Johari (Data Engineer)	Darren Tang (YCEP Coordinator)
School of Computer and Mathematical Sciences	Cube Asia	Y Suite Waymouth
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